

3-1 Tangent Lines and the Derivative at a Point

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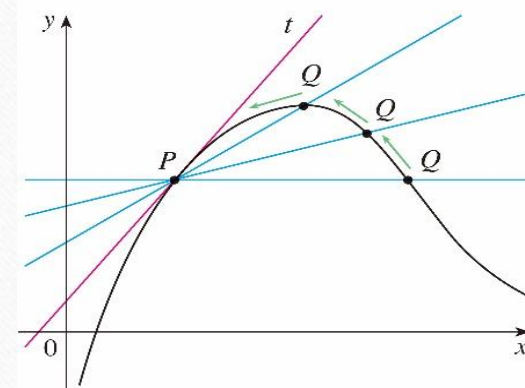
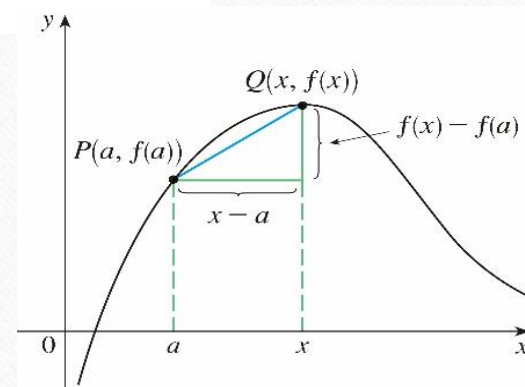
師大工教一

※Tangent

Recall: slope _{斜率} → slope of the secant → slope of the tangent

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$
$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$x - a = h$$
$$x = h + a$$



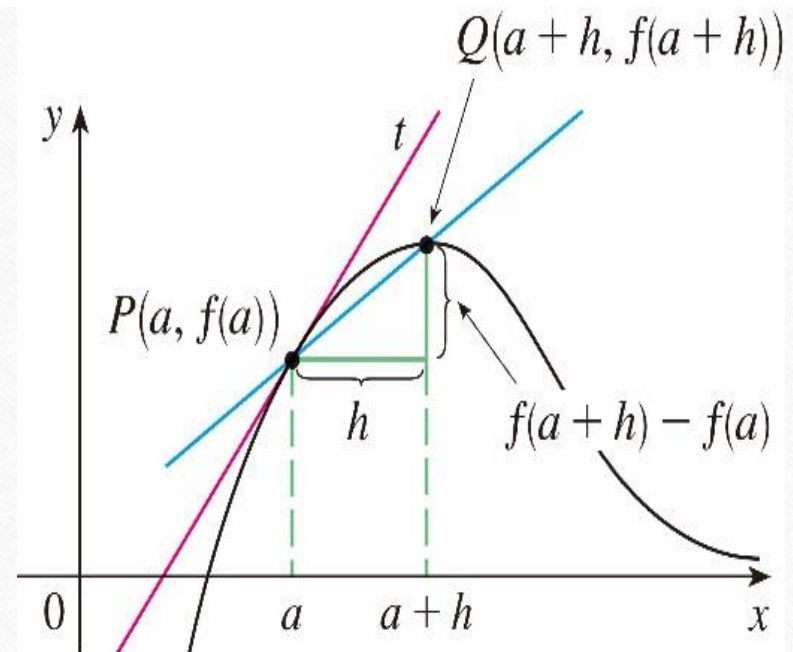
Definition: The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$

with slope $m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ provided that the limit exists.

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad x - a = h \quad x = a + h$$

$$m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$



Ex1(p135)

(a) Find the slope of the curve $y = \frac{1}{x}$ at any point $x = a \neq 0$. What is the slope

at the point $x = 1$?

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a} = \lim_{x \rightarrow a} \frac{\frac{a - x}{ax}}{x - a} = \lim_{x \rightarrow a} \frac{-1}{ax} = -\frac{1}{a^2}$$

(b) Where does the slope equal $-\frac{1}{4}$? $-\frac{1}{a^2} = -\frac{1}{4}$ $a = \pm 2$

$$At (2, \frac{1}{2}), (-2, -\frac{1}{2}), \dots$$

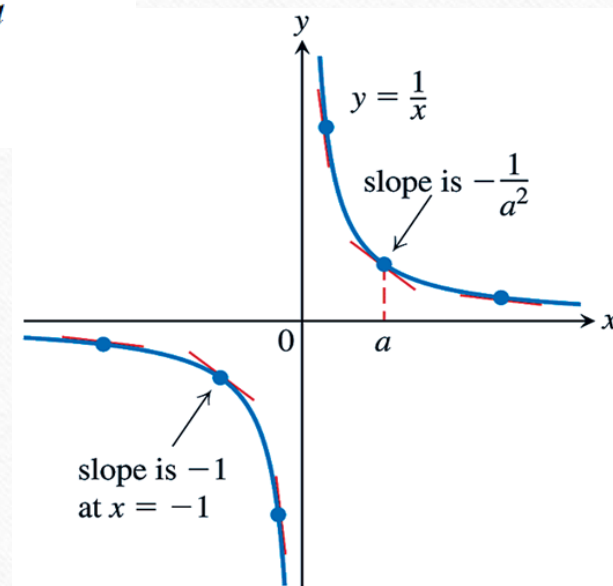
(c) What happens to the tangent line to the curve at the point $(a, \frac{1}{a})$ as a

changes?

$$m_{tan} \Big|_{x=a} = -\frac{1}{a^2}$$

$$-\infty \rightarrow -1 \rightarrow 0^- \quad 0^+ \rightarrow 1 \rightarrow \infty$$

$$0 \rightarrow -1 \rightarrow -\infty \quad -\infty \rightarrow -1 \rightarrow 0$$



※Rates of Change: Derivative at a Point

Definiton: The **derivative** of f at a point a , denoted by $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided this limit exists. As the above limit exist at $x = a$, we say that function f is **differentiable** at $x = a$.
可微分

Ex: Let $f(x) = \frac{(x-1)(x-2)(x-3)}{(x-4)(x-5)}$, find $f'(3)$.

$$f'(3) = \lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} = \frac{(x-1)(x-2)(x-3) - 0}{(x-4)(x-5)} = \lim_{x \rightarrow 3} \frac{(x-1)(x-2)}{(x-4)(x-5)} = \frac{2}{2} = 1$$

-1×-2

HW3-1

- HW: 6,7,9,13,23,24,33,34

3. (a) Write the definition of the derivative $f'(x_0)$ of a function f at a point x_0 . (2 pts)

(b) Show that $f(x) = |x|$ is continuous but not differentiable at $x_0 = 0$. (8 pts)

$$(a) f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

(b) (1) f is conti. at $x=0$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} (-x) = 0$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} |x| = 0$$

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(2) claim: f is not differentiable at $x_0 = 0$

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{|x| - |0|}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1$$

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{|x| - |0|}{x - 0} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1$$

$$\therefore \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \text{ DNE.}$$

$\Rightarrow f$ is not differentiable at $x=0$.

3. (a) Write the definition of the derivative $f'(x_0)$ of a function f at a point x_0 . (2 pts)
- (b) Show that $f(x) = |x|$ is continuous but not differentiable at $x_0 = 0$. (8 pts)

3-2 The Derivative as a Function

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Definition: The **derivative** of the function $f(x)$ is $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

provided the limit exists. The process of calculating a derivative is called **differentiation**.

Ex1(p139) Differentiate $f(x) = \frac{x}{x-1}$.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{x+h}{x+h-1} - \frac{x}{x-1}}{h} = \lim_{h \rightarrow 0} \frac{\frac{(x+h)(x-1) - x(x+h-1)}{(x+h-1)(x-1)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{(x+h)}\cancel{(x-1)} - x\cancel{(x-1)} - xh}{(\cancel{x+h-1})(\cancel{x-1})} = \lim_{h \rightarrow 0} \frac{x+h-x-xh}{x+h-1} \times \frac{1}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(x-1) - xh}{h(x-1+h)(x-1)} = \lim_{h \rightarrow 0} \frac{-1}{(x-1+h)(x-1)} = \frac{-1}{(x-1)^2} \end{aligned}$$

Ex2(p139)

(a) Find the derivative of $f(x) = \sqrt{x}$ for $x > 0$.

(b) Find the tangent line to the curve $f(x) = \sqrt{x}$ at $x = 4$.

$$(a) f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} = \lim_{h \rightarrow 0} \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{1}{(\sqrt{x+h} + \sqrt{x})} = \frac{1}{2\sqrt{x}}$$

(b) 點斜式, 過 (x_0, y_0) , 斜率 m 直線方程式: $y - y_0 = m(x - x_0)$
 $x = 4$ $f(4) = \sqrt{4} = 2$, 過 $(4, 2)$ $\therefore$ Eq. of tangent line.

$$m_{\text{tan}} = f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

$$y - 2 = \frac{1}{4}(x - 4)$$

$$+y - 8 = x - 4$$

$$x - 4y + 4 = 0.$$

Notation: $f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = D(f)(x) = D_x f(x)$

$$f'(a) = \left. \frac{dy}{dx} \right|_{x=a} = \left. \frac{df}{dx} \right|_{x=a} = \left. \frac{d}{dx} f(x) \right|_{x=a}$$

A function $y = f(x)$ is **differentiable on an open interval** (finite or infinite) if it has a derivative at each point of the interval. It is **differentiable on a closed interval** $[a, b]$ if it is differentiable on the interval (a, b) and if the limit

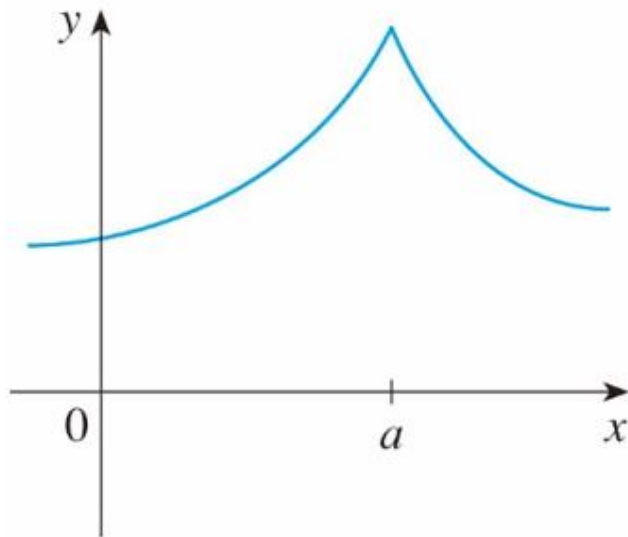
$$\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} \quad \text{Right-hand derivative at } a$$

$$\lim_{h \rightarrow 0^-} \frac{f(b+h) - f(b)}{h} \quad \text{Left-hand derivative at } b$$

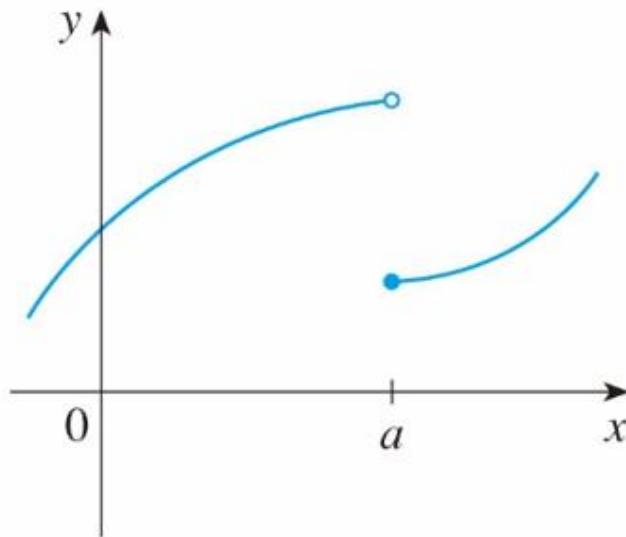
Ex4(p141) Show that the function $f(x) = |x|$ is differentiable on $(-\infty, 0)$ and on $(0, \infty)$ but has no derivative at $x = 0$.

Note: A differentiable function is **smooth**.

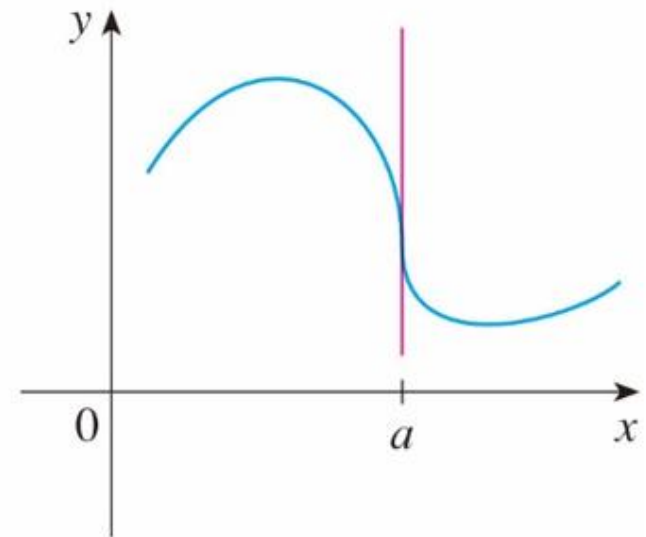
When does a function not have a derivative at a point?



(a) A corner



(b) A discontinuity



(c) A vertical tangent

Theorem 1-Differentiability Implies Continuity If f has a derivative at $x=c$, then f is continuous at $x=c$.

$\therefore f$ is conti. at $x=c$.

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} \cdot (x - c)$$

$$f(x) = f(c) + \frac{f(x) - f(c)}{x - c} \cdot (x - c)$$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} f(c) + \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \cdot \lim_{x \rightarrow c} (x - c)$$

$$\lim_{x \rightarrow c} f(x) = f(c) + f'(c) \cdot 0 = f(c)$$

Caution: The converse of Theorem 1 is false. A function need not have a derivative at a point where it is continuous.

HW3-2

- HW: 2,4,8,9,27~30,38,39,59

3-3 Differentiation Rules

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$$1. \frac{d}{dx}(c) = 0$$

$$2. \frac{d}{dx}(x) = 1$$

$$3. n \text{ is any real number, } \frac{d}{dx}(x^n) = n x^{n-1}$$

Ex1(p148): Differentiate the following powers of x .

$$(a) x^3$$

$$\Downarrow \\ 3x^2$$

$$(b) x^{2/3}$$

$$(c) x^{\sqrt{2}}$$

$$(d) \frac{1}{x^4}$$

$$\rightarrow -4x^{-5}$$

$$(e) x^{-4/3}$$

$$\rightarrow \frac{4}{3} x^{-7/3}$$

$$(f) \sqrt{x^{2+\pi}}$$

$$\left(x^{2+\pi} \right)^{1/2}$$

$$= x^{1+\frac{\pi}{2}}$$

$$\Downarrow \\ \frac{2+\pi}{2} x^{\frac{\pi}{2}}$$

微分的线性性质

$$4. \frac{d}{dx}[cu] = c \frac{du}{dx}$$

$$5. \frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$$

Ex3(p150) Find the derivative of the polynomial $y = x^3 + \frac{4}{3}x^2 - 5x + 1$

$$\frac{dy}{dx} = 3x^2 + \frac{8}{3}x - 5$$

$$6. \frac{d}{dx}(e^x) = e^x$$

$$7. \text{The Product Rule: } \frac{d}{dx}(uv) = \frac{du}{dx} \cdot v + u \cdot \frac{dv}{dx}$$

Ex6(p152): Find the derivative of (a) $y = \frac{1}{x}(x^2 + e^x)$ (b) $y = e^{2x} = (e^x)^2$

$$(a) \Rightarrow -x^{-2}(x^2 + e^x) + \frac{1}{x}(2x + e^x)$$

$$= -1 - x^{-2}e^x + 2 + \frac{e^x}{x}$$

$$= \frac{e^x}{x} - x^{-2}e^x + 1$$

$$(b) \frac{dy}{dx} = 2e^x$$

8. The Quotient Rule: $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{\frac{du}{dx} \cdot v - u \cdot \frac{dv}{dx}}{v^2}$

Ex7(p153) Find the derivative of (a) $y = \frac{t^2 - 1}{t^3 + 1}$ (b) $y = e^{-x}$

$$\begin{aligned} \text{(a)} \quad & \frac{2t(t^3 + 1) - (t^2 - 1)3t^2}{(t^3 + 1)^2} \\ &= \frac{2t^4 + 2t - 3t^4 + 3t^2}{t^6 + 2t^3 + 1} = \frac{-t^4 + 3t^2 + 2t}{t^6 + 2t^3 + 1} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad f'(x) &= \frac{\frac{1}{e^x} \cdot (-1) \cdot e^x}{e^{2x}} \\ &= \frac{-1}{e^x} \end{aligned}$$

Ex8(p154) Find the derivative of $y = \frac{(x-1)(x^2-2x)}{x^4}$.

(cf: use quotient rule or not)

$$\begin{aligned} \frac{dy}{dx} &= \frac{(2x-3)(\cancel{x^3}) - (x^2-3x+2)(\cancel{3x^3})}{x^{\cancel{4}^4}} = \frac{(x-1)(x-2)}{x^3} = \frac{x^2-3x+2}{x^3} = \frac{1}{x} - \frac{3}{x^2} + \frac{2}{x^3} \\ &= \frac{2x^2-3x-3x^2+9x-6}{x^4} = \frac{-x^2+6x-6}{x^4} \end{aligned}$$

Higher-Order Derivatives

Newton: $f(x) \rightarrow f'(x) \rightarrow f''(x) \rightarrow f'''(x) \rightarrow f^{(4)}(x) \rightarrow \cdots \rightarrow f^{(n)}(x) \rightarrow \cdots$

Leibniz: $y \rightarrow \frac{dy}{dx} \rightarrow \frac{d^2y}{dx^2} \rightarrow \frac{d^3y}{dx^3} \rightarrow \cdots \rightarrow \frac{d^ny}{dx^n} \rightarrow \cdots$

HW3-3

- HW:3,10,15,18,30,45,53,57,69

3-5 Derivatives of Trigonometric Functions

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$$1. \frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

Ex1(p167): Find the derivatives of the following functions:

$$(a) y = x^2 - \sin x \quad (b) y = e^x \sin x \quad (c) y = \frac{\sin x}{x}$$

$$y' = 2x - \cos x$$

$$y' = e^x \sin x + e^x \cos x$$

$$y' = \frac{\cos x \cdot x - \sin x}{x^2}$$

$$2. \frac{d}{dx}(\cos x) = -\sin x$$

Ex2(p167): Find the derivatives of the following functions:

(a) $y = 5e^x + \cos x$ (b) $y = \sin x \cos x$ (c) $y = \frac{\cos x}{1 - \sin x}$

$$y' = 5e^x - \sin x$$

$$y' = \cos^2 x - \sin^2 x \\ = \cos 2x$$

$$y' = \frac{-\sin x(1 - \sin x) + \cos^2 x}{(1 - \sin x)^2} = \frac{-\sin x + \sin^2 x + \cos^2 x}{(1 - \sin x)^2} \\ = \frac{1 - \sin x}{(1 - \sin x)^2} = \frac{1}{1 - \sin x} \quad \ast$$

$$3. \frac{d}{dx}(\tan x) = \sec^2 x$$

$$4. \frac{d}{dx}(\cot x) = -\csc^2 x$$

$$5. \frac{d}{dx}(\sec x) = \sec x \cdot \tan x$$

$$6. \frac{d}{dx}(\csc x) = -\csc x \cdot \cot x$$

Ex5(p169): Find $\frac{d}{dx}(\tan x)$.

$$\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

Ex6(p170): Find $\lim_{x \rightarrow 0} \frac{\sqrt{2 + \sec x}}{\cos(\pi - \tan x)}$.

$$\lim_{x \rightarrow 0} \frac{\sqrt{2 + \sec x}}{\cos(\pi - \tan x)} = \frac{\sqrt{3}}{\cos \pi} = \frac{\sqrt{3}}{-1} = -\sqrt{3}$$

HW3-5

- HW: 3,6,7,12,17,34,39,52,53

3-6 The Chain Rule

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The Chain Rule If $f(u)$ is differentiable at $u = g(x)$ and $g(x)$ is differentiable at x , then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x and $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$. In Leibniz notation, if $y = f(u)$ and $u = g(x)$, then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$, where $\frac{dy}{du}$ is evaluated at $u = g(x)$.

Ex1(p173): Differentiate $y = (3x^2 + 1)^2$.

$$y' = 2(3x^2 + 1)(6x) = 12x^3 + 12x$$

Formula: (1) Power Chain Rule: $\left[(f(x))^n \right]' = n \cdot (f(x))^{n-1} \cdot f'(x)$

$$(2) \left[\sin(f(x)) \right]' = \cos(f(x)) \cdot f'(x)$$

$$\left[\cos(f(x)) \right]' = -\sin(f(x)) \cdot f'(x)$$

$$\left[\tan(f(x)) \right]' = \sec^2(f(x)) \cdot f'(x)$$

$$\left[\cot(f(x)) \right]' = -\csc^2(f(x)) \cdot f'(x)$$

$$\left[\sec(f(x)) \right]' = \sec(f(x)) \cdot \tan(f(x)) \cdot f'(x)$$

$$\left[\csc(f(x)) \right]' = -\csc(f(x)) \cdot \cot(f(x)) \cdot f'(x)$$

Ex3(p175) Differentiate $y = \sin(x^2 + e^x)$.

$$y' = (2x + e^x) \cos(x^2 + e^x)$$

Formula: (3) $\left[e^{f(x)} \right]' = e^{f(x)} \cdot f'(x)$

Ex4(p175) Differentiate $y = e^{\cos x}$.

$$y' = e^{\cos x} (-\sin x) = -e^{\cos x} \sin x$$

Repeated Use of the Chain Rule

Ex: Differentiate $y = \cos \sqrt{\sin(\tan \pi x)} \cdot [\sin(\tan \pi x)]^{\frac{1}{2}}$

$$y' = -\sin(\sin(\tan \pi x))^{\frac{1}{2}} \cdot \frac{1}{2} (\sin(\tan \pi x))^{-\frac{1}{2}} \cdot \cos(\tan \pi x) \cdot (\tan \pi x)'$$

$$= \frac{-\sin \sqrt{\sin(\tan \pi x)}}{\sqrt{\sin(\tan \pi x)}} \cdot \cos(\tan \pi x) \cdot \sec^2(\pi x) \cdot \pi$$

$$= \frac{\pi \sec^2(\pi x) \cdot \cos(\tan \pi x) \cdot (-\sin \sqrt{\sin(\tan \pi x)})}{\sqrt{\sin(\tan \pi x)}}$$

Ex6(p176) The Power Chain Rule simplifies computing the derivative of a power of an expression.

$$(a) \frac{d}{dx} (5x^3 - x^4)^7$$

$$= 7(5x^3 - x^4)^6 \cdot (15x^2 - 4x^3)$$

$$(b) \frac{d}{dx} \left(\frac{1}{3x-2} \right)$$

$$= \frac{-3}{(3x-2)^2}$$

$$(c) \frac{d}{dx} (\sin^5 x)$$

$$= \frac{d}{dx} (\sin x)^5$$

$$= 5(\sin^4 x) \cdot \cos x$$

$$(d) \frac{d}{dx} (e^{\sqrt{3x+1}})$$

$$= e^{\sqrt{3x+1}} \cdot \left[(\sqrt{3x+1})' \right]$$

$$= e^{\sqrt{3x+1}} \cdot \frac{1}{2} (3x+1)^{-\frac{1}{2}} \cdot 3$$

$$= \frac{3e^{\sqrt{3x+1}}}{2\sqrt{3x+1}}$$

HW3-6

- HW: 10,14,17,21,33,38,43,48,61,74,87

104 (d) (5 points) $y = \sqrt{2 + \sqrt{2 + \sqrt{x}}}$

$$\begin{aligned} \frac{d}{dx}(\sqrt{2 + \sqrt{2 + \sqrt{x}}}) &= \frac{d}{dx} \left(2 + (2 + x^{\frac{1}{2}})^{\frac{1}{2}} \right)^{\frac{1}{2}} \\ &= \frac{1}{2} \left(2 + (2 + x^{\frac{1}{2}})^{\frac{1}{2}} \right)^{-\frac{1}{2}} \cdot \frac{1}{2} (2 + x^{\frac{1}{2}})^{-\frac{1}{2}} \cdot \frac{1}{2} x^{-\frac{1}{2}} \\ &= \frac{1}{8} \cdot \frac{1}{\sqrt{2 + \sqrt{2 + \sqrt{x}}}} \cdot \frac{1}{\sqrt{2 + \sqrt{x}}} \cdot \frac{1}{\sqrt{x}} \\ &= \frac{1}{8} \cdot \frac{1}{\sqrt{(2 + \sqrt{2 + \sqrt{x}})(2 + \sqrt{x})(x)}} = \frac{1}{8\sqrt{(2 + \sqrt{2 + \sqrt{x}})(2 + \sqrt{x})(x)}} \end{aligned}$$

$$y = e^{3x} \sin(\cos^2(3x))$$

$$\begin{aligned} \frac{d}{dx}(e^{3x} \sin(\cos^2(3x))) &= (e^{3x})'(\sin(\cos^2(3x))) + (e^{3x})(\sin(\cos^2(3x)))' \\ &= 3e^{3x}(\sin(\cos^2(3x))) + e^{3x}(\cos(\cos^2(3x))) \left[(\cos^2(3x))' \right] \\ &= 3e^{3x}(\sin(\cos^2(3x))) + e^{3x}(\cos(\cos^2(3x))) (2(\cos(3x))(-\sin 3x)) \\ &= 3e^{3x}(\sin(\cos^2(3x))) - 6e^{3x}(\cos(\cos^2(3x)))(\cos 3x \sin 3x) \end{aligned}$$

隱微分法

3-7 Implicit Differentiation

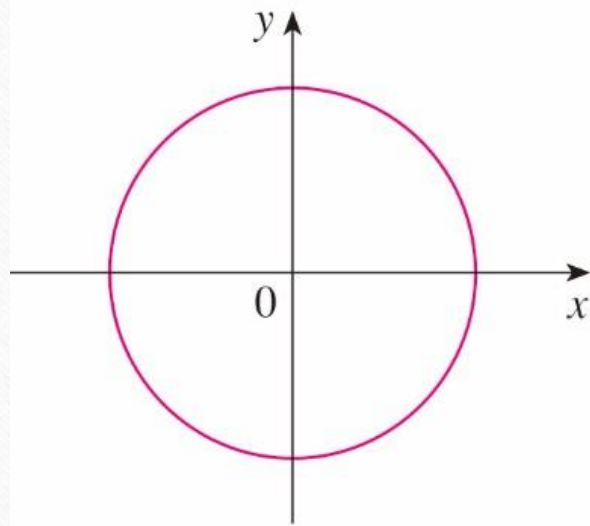
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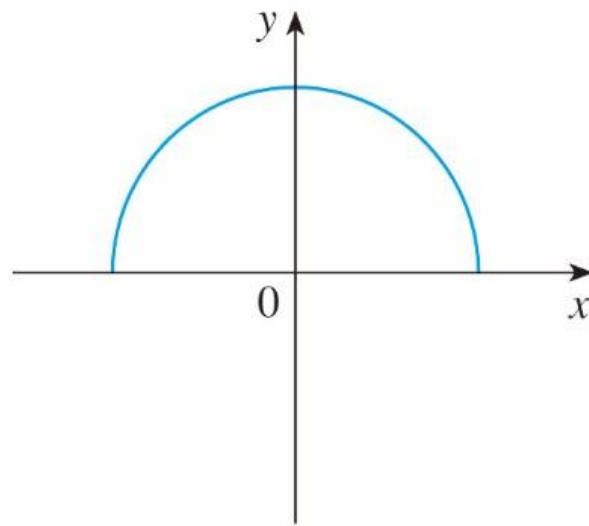
師大工教一

Explicit functions: $y = \sqrt{x^2 + 1}$ or $y = x \sin x$

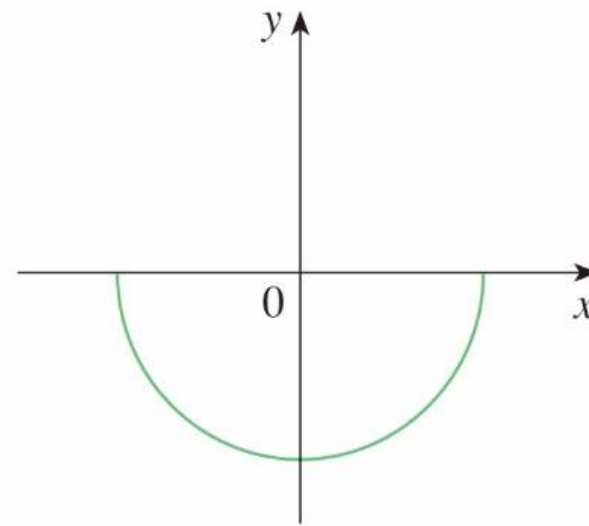
Implicit functions: $x^2 + y^2 = 25$



(a) $x^2 + y^2 = 25$



(b) $f(x) = \sqrt{25 - x^2}$

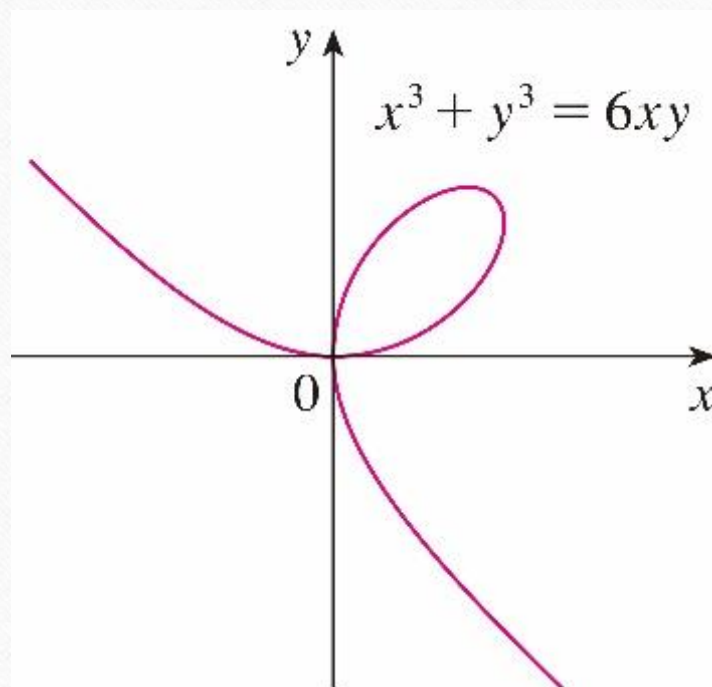


(c) $g(x) = -\sqrt{25 - x^2}$

笛卡兒葉形線

Folium of Descartes: $x^3 + y^3 = 6xy$

笛卡兒



Implicit Differentiation

1. Differentiate both sides of the equation with respect to x , treating y as a differentiable function of x .
2. Collect the terms with $\frac{dy}{dx}$ on one side of the equation and solve for $\frac{dy}{dx}$.

Ex2(p181) Find the slope of the circle $x^2 + y^2 = 25$ at the point $(3, -4)$.

$$y = \pm \sqrt{25 - x^2}$$

the point $(3, -4)$ is on lower semicircle

so differentiate $y = -\sqrt{25 - x^2}$

$$y' = \frac{x}{\sqrt{25 - x^2}}$$

By Implicit Differentiation.

$$x^2 + y^2 = 25$$

$$2x + 2y y' = 0$$

$$y y' = -x$$

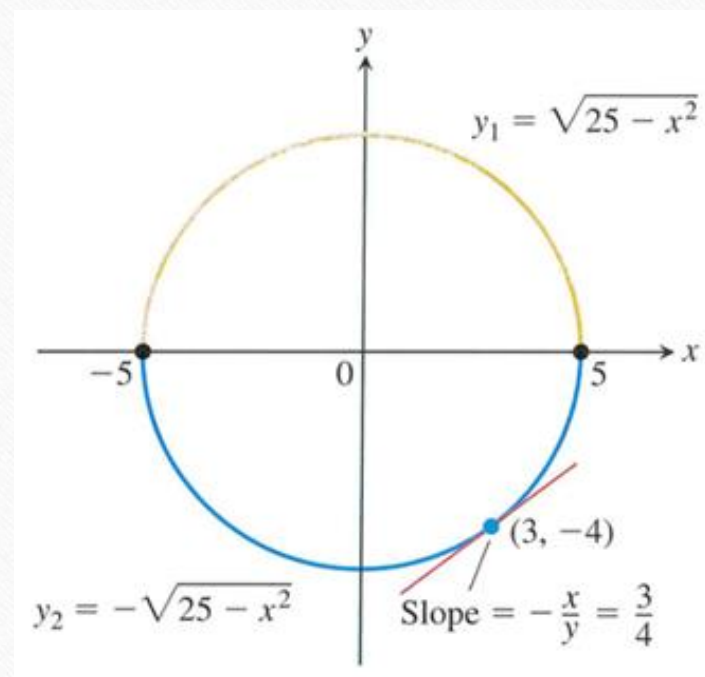
$$y' = \frac{-x}{y}$$

By Implicit Differentiation.

$$x^2 + y^2 = 25$$
$$2x + 2y y' = 0$$
$$y y' = -x$$
$$y' = \frac{-x}{y}$$
$$y'(3, -4) = \frac{-3}{-4} = \frac{3}{4}$$

$x^2 + (f(x))^2 = 25$

$$2x + 2(f(x)) \cdot f'(x) = 0$$
$$2f(x) \cdot f'(x) = -2x$$
$$f'(x) = \frac{-2x}{2f(x)}$$



Ex3(p182) Find $\frac{dy}{dx}$ if $y^2 = x^2 + \sin xy$.

$$x^2 - y^2 + \sin xy = 0$$

$$2x - 2yy' + \cos xy (y + xy') = 0$$

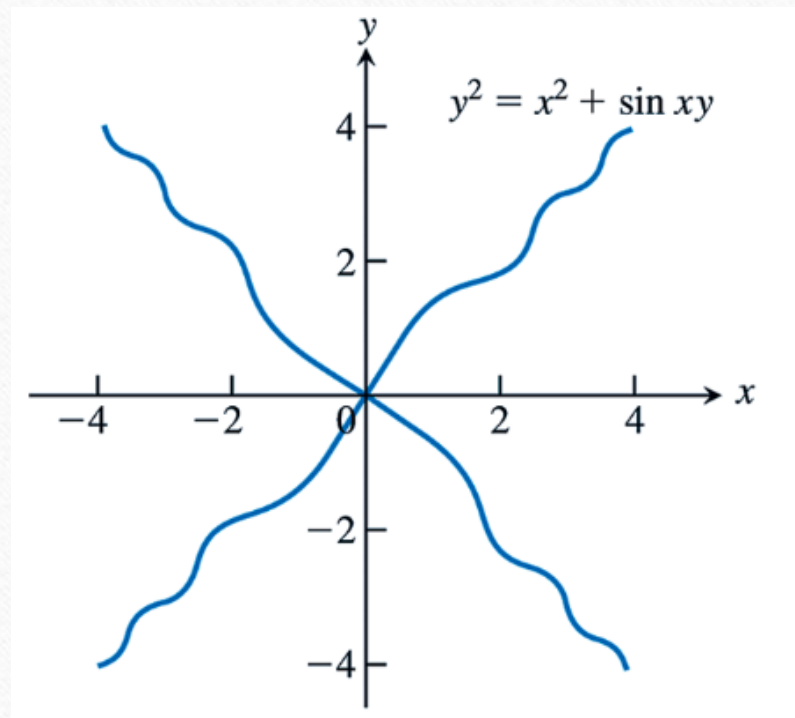
$$2yy' = 2x + \cos xy (y + xy')$$

$$2yy' = 2x + y \cos xy + xy' \cos xy$$

$$2yy' - xy' \cos xy = 2x + y \cos xy$$

$$y'(2y - x \cos xy) = 2x + y \cos xy$$

$$y' = \frac{2x + y \cos xy}{2y - x \cos xy}$$



Derivatives of Higher Order

Ex4(p182) Find $\frac{d^2y}{dx^2}$ if $2x^3 - 3y^2 = 8$.

$$3y^2 = 2x^3 - 8$$

$$6yy' = 6x^2$$

$$y' = \frac{x^2}{y}$$

$$\begin{aligned} y'' \frac{d}{dx} \left(\frac{x^2}{y} \right) &= \frac{2xy - x^2 y'}{y^2} = \frac{2xy - x^2 \cdot \frac{x^2}{y}}{y^2} \\ &= \frac{2xy - \frac{x^4}{y}}{y^2} = \frac{2xy^2 - x^4}{y^3} \end{aligned}$$

Ex5(p183) Show that the point $(2,4)$ lies on the curve $x^3 + y^3 - 9xy = 0$. Then find the tangent line and normal line to the curve there.

$$x^3 + y^3 - 9xy = 0 \bigg|_{(2,4)} = 8 + 64 - 72 = 0$$

$$\therefore (2,4) \text{ lies on } x^3 + y^3 - 9xy = 0$$

$$3x^2 + 3y^2y' - 9(y + xy') = 0$$

$$3x^2 + 3y^2y' - 9y - 9xy' = 0$$

$$3y^2y' - 9xy' = 9y - 3x^2$$

$$y'(3y^2 - 9x) = 9y - 3x^2$$

$$y' = \frac{-3x^2 + 9y}{3y^2 - 9x}$$

$$y' \big|_{(2,4)} = \frac{-12 + 36}{48 - 18} = \frac{24}{30} = \frac{4}{5} = m_{\text{tan.}}$$

Eg. of tangent line:

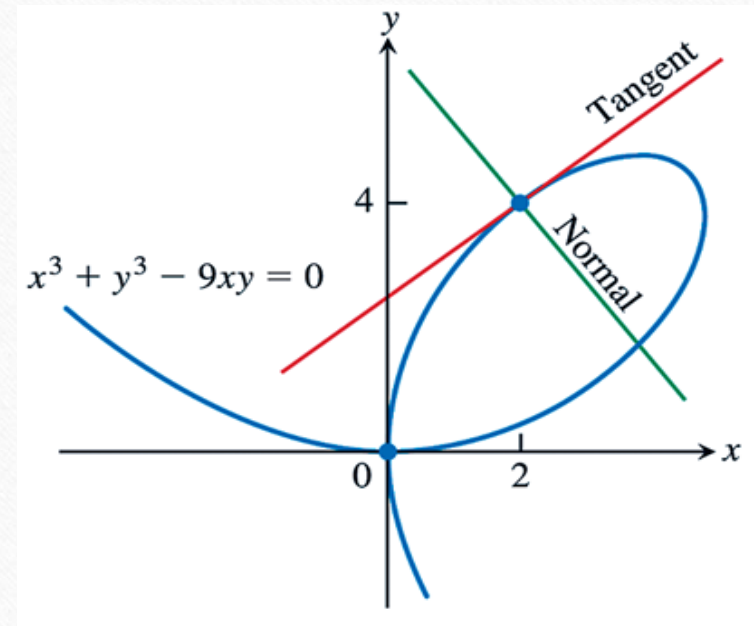
$$y - 4 = \frac{4}{5}(x - 2)$$

$$4x - 5y + 12 = 0$$

Eg. of normal line:

$$y - 4 = -\frac{4}{5}(x - 2)$$

$$4x + 5y - 28 = 0$$



HW3-7

- HW: 1,2,10,26,35,38

105分

4. (10 points) Find an equation of the tangent line to the graph at the given point (x_0, y_0)

$$x^2 + \sqrt{xy} + y^2 = 72, \quad (x_0, y_0) = (2, 8).$$

$$2x + \frac{1}{2\sqrt{xy}}(y + xy') + 2yy' = 0$$

$$2x + \frac{y}{2\sqrt{xy}} + \frac{xy'}{2\sqrt{xy}} + 2yy' = 0$$

$$y'(\frac{x}{2\sqrt{xy}} + 2y) = -2x - \frac{y}{2\sqrt{xy}}$$

$$y' = \frac{-2x - \frac{y}{2\sqrt{xy}}}{\frac{x}{2\sqrt{xy}} + 2y}$$

$$y'(2, 8) = \frac{-4 - \frac{8}{8}}{\frac{1}{4} + 16} = \frac{-5}{\frac{65}{4}} = \frac{-4}{13}$$

Eq. of tangent line:

$$y - 8 = \frac{-4}{13}(x - 2) \rightarrow 4x + 13y - 112 = 0$$

110 5. Given $x^3 + ye^x + y^3 = 2$. Find the tangent line at point $P = (0, 1)$. (6 pts)

$$3x^2 + y'e^x + ye^x + 3y^2y' = 0$$

$$y'(e^x + 3y^2) = -3x^2 - ye^x$$

$$y' = \frac{-3x^2 - ye^x}{e^x + 3y^2}$$

$$y'(0, 1) = \frac{-1}{4}$$

Eq. of tangent line:

$$y - 1 = \frac{-1}{4}x$$

$$x + 4y - 4 = 0$$

3-8 Derivatives of Inverse Functions and Logarithms

師大工教一

Ex: $f(x) = \frac{1}{2}x + 1$

$$\frac{d}{dx} f(x) = \frac{1}{2}$$

$$f^{-1}(x) = 2x - 2$$

$$\frac{d}{dx} f^{-1}(x) = 2$$

Theorem 3-The Derivative Rule for Inverses

If f has an interval I as domain and $f'(x)$ exists and is never zero on I ,

then f^{-1} is differentiable at every point in its domain (the range of f). The

value of $(f^{-1})'$ at a point b in the domain of f^{-1} is the reciprocal of the

value of f' at the point $a = f^{-1}(b)$: $(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$ or

$$\left. \frac{df^{-1}}{dx} \right|_{x=b} = \frac{1}{\left. \frac{df}{dx} \right|_{x=f^{-1}(b)}}.$$

$$\begin{aligned} & \left\{ \begin{aligned} f(f^{-1}(x)) &= x \\ \rightarrow f'(f^{-1}(x)) \cdot (f^{-1})'(x) &= 1 \\ (f^{-1})'(x) &= \frac{1}{f'(f^{-1}(x))} \end{aligned} \right. \end{aligned}$$

Ex2(p187) Let $f(x) = x^3 - 2, x > 0$. Find the value of $\frac{df^{-1}}{dx}$ at $x = 6 = f(2)$

without finding a formula for $f^{-1}(x)$.

$$\frac{df^{-1}}{dx} = \frac{1}{f'(f^{-1}(b))} = \frac{1}{12}$$

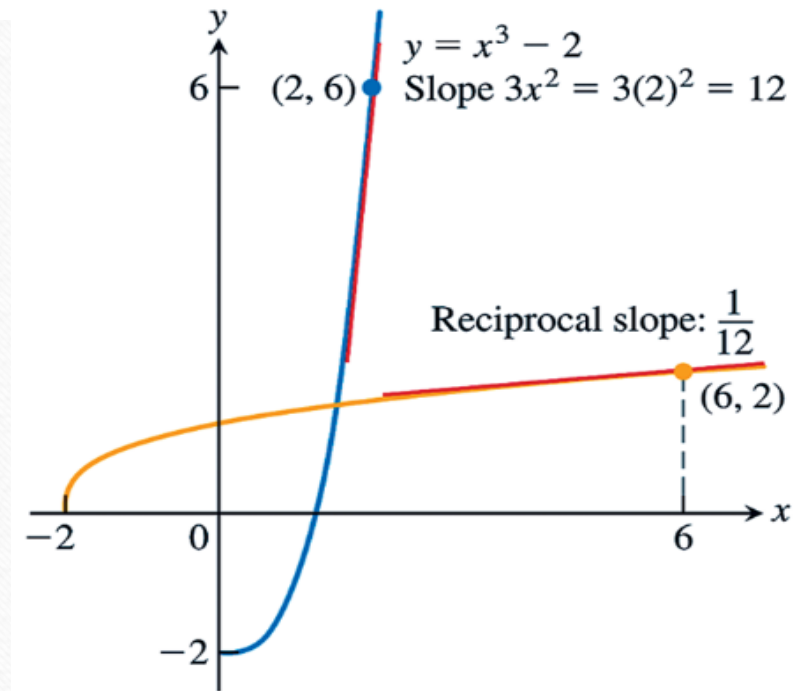
$$b = f(2)$$

$$f^{-1}(b) = f^{-1}(f(2)) = 2$$

$$f(x) = x^3 - 2$$

$$f'(x) = 3x^2$$

$$f'(2) = 3 \times 2^2 = 12$$



Derivative of the Natural Logarithm Function

Formula: 1. $\frac{d}{dx}(\ln x) = \frac{1}{x}$ 2. $\frac{d}{dx}(\ln|x|) = \frac{1}{x}$ 3. $\frac{d}{dx}(\ln f(x)) = \frac{f'(x)}{f(x)}$

$$e^{\ln x} = x \quad (\ln x)' = \frac{1}{e^{\ln x}} = \frac{1}{x}$$

$$e^{\ln x} \cdot (\ln x)' = 1$$

$$e^{\ln f(x)} = f(x)$$

$$e^{\ln f(x)} \cdot (\ln f(x))' = f'(x)$$

$$(\ln f(x))' = \frac{f'(x)}{f(x)}$$

Ex3(p188) Find (1) $\frac{d}{dx}(\ln 2x)$ (2) $\frac{d}{dx}(\ln(x^2 + 3))$

$$= \frac{1}{2x} \cdot 2 = \frac{1}{x}$$

$$= \frac{2x}{x^2 + 3}$$

The Derivatives of a^u and $\log_a u$

Formula: 4. $\frac{d}{dx}(a^x) = a^x \ln a$

$$\begin{aligned}\frac{d}{dx}(a^x) &= \frac{d}{dx}(e^{\ln a^x}) \\ &= \frac{d}{dx}(e^{\ln a \cdot x}) \\ &= e^{\ln a \cdot x} \cdot (\ln a \cdot x)' \\ &= a^x \ln a\end{aligned}$$

5. $\frac{d}{dx}(a^u) = a^u \cdot \ln a \cdot \frac{du}{dx}$ (or $(a^{f(x)})' = a^{f(x)} \cdot \ln a \cdot f'(x)$)

Ex5(p190) Find (a) $\frac{d}{dx}(3^x)$ (b) $\frac{d}{dx}(3^{-x})$ (c) $\frac{d}{dx}(3^{\sin x})$

$$= 3^x (\ln 3)$$

$$\begin{aligned}&= \frac{d}{dx}(e^{\ln 3 \cdot (-x)}) \\ &= e^{\ln 3 \cdot (-x)} \cdot (-\ln 3) \\ &= 3^{-x} (-\ln 3)\end{aligned}$$

$$\begin{aligned}&= \frac{d}{dx}(e^{\ln 3 \cdot \sin x}) \\ &= e^{\ln 3 \cdot \sin x} \cdot (\ln 3 \cdot \cos x) \\ &= 3^{\sin x} \cdot \ln 3 \cdot \cos x\end{aligned}$$

Formula: 6. $\frac{d}{dx}(\log_a x) = \frac{1}{\ln a \cdot x}$ 5. $\frac{d}{dx}(\log_a u) = \frac{1}{u \ln a} \cdot \frac{du}{dx}$

(or $(\log_a(f(x)))' = \frac{1}{f(x) \cdot \ln a} \cdot f'(x)$)

$\frac{d}{dx}(\log_a x) = \frac{d}{dx}\left(\frac{\log_e x}{\log_e a}\right) = \frac{d}{dx}\left(\frac{\ln x}{\ln a}\right) = \frac{1}{\ln a} \frac{d}{dx}(\ln x) = \frac{1}{\ln a} \cdot \frac{1}{x}$

$(e^{\ln(x^2+1)})' = e^{\ln(x^2+1)} \cdot (\ln(x^2+1))' = 2x$
 $(\ln(x^2+1))' = \frac{2x}{x^2+1}$

Logarithmic Differentiation

Ex6(p191) Find $\frac{dy}{dx}$ if $y = \frac{(x^2+1)(x+3)^{\frac{1}{2}}}{x-1}, x > 1$

$\ln y = \ln\left(\frac{(x^2+1)(x+3)^{\frac{1}{2}}}{x-1}\right) = \ln(x^2+1) + \ln(x+3)^{\frac{1}{2}} - \ln(x-1)$
 $= \ln(x^2+1) + \frac{1}{2}\ln(x+3) - \ln(x-1)$

$\frac{1}{y} \cdot y' = \frac{2x}{x^2+1} + \frac{1}{2} \cdot \frac{1}{x+3} - \frac{1}{x-1}$

$= \frac{2x}{x^2+1} + \frac{1}{2(x+3)} - \frac{1}{x-1}$

$y' = \frac{(x^2+1)(x+3)^{\frac{1}{2}}}{x-1} \left[\frac{2x}{x^2+1} + \frac{1}{2(x+3)} - \frac{1}{x-1} \right]$

Ex7(p193) Differentiate $f(x) = x^x, x > 0$.

$$\ln f(x) = x \cdot \ln x$$

$$\frac{f'(x)}{f(x)} = \ln x + x \cdot \frac{1}{x} = \ln x + 1$$

$$f'(x) = x^x (\ln x + 1)$$

$$\ln f(x) = \ln(x^x) = x \ln x$$

$$(e^{\ln f(x)})' = (f(x))'$$

$$f(x) \cdot (\ln f(x))' = (f(x))'$$

$$(\ln f(x))' = \frac{(f(x))'}{f(x)}$$

$$\frac{f'(x)}{f(x)} = (x \ln x)' = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1$$

$$f'(x) = x^x (\ln x + 1)$$

Theorem 4-The Number e as a Limit The number e can be calculated as

the limit $e = \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$. (L'Hopital's Rule)

HW3-8

- **HW: 4,7,20,24,37,42,48,52,64,95**

107

6. (10 pts) Use the Logarithmic Differentiation (對數微分法) to find an equation of the tangent line (切線方程式) to the graph of $y = \frac{(x+2)^2}{\sqrt{x^2+1}}$ at the point (0, 4).

$$\ln y = \ln \left(\frac{(x+2)^2}{\sqrt{x^2+1}} \right) = 2 \ln(x+2) - \frac{1}{2} \ln(x^2+1)$$

$$\frac{y'}{y} = 2 \left(\frac{1}{x+2} \right) - \frac{1}{2} \left(\frac{2x}{x^2+1} \right) = \frac{2}{x+2} - \frac{x}{x^2+1}$$

$$y' = \frac{(x+2)^2}{\sqrt{x^2+1}} \left(\frac{2}{x+2} - \frac{x}{x^2+1} \right)$$

$$y' \Big|_{x=0} = \frac{4}{1} (1-0) = 4$$

102 2 (f) (10 points) $y = \frac{(x+2)^5(e^x-1)^2}{x^3(2x-1)^3}$

$$\ln y = 5 \ln(x+2) + 2 \ln(e^x-1) - 3 \ln(2x^2-x)$$

$$\frac{y'}{y} = 5 \left(\frac{1}{x+2} \right) + 2 \left(\frac{e^x}{e^x-1} \right) - 3 \left(\frac{4x-1}{2x^2-x} \right)$$

$$y' = \frac{(x+2)^5(e^x-1)^2}{x^3(2x-1)^3} \left(\frac{5}{x+2} + \frac{2e^x}{e^x-1} - \frac{12x-3}{2x^2-x} \right)$$

$$e^{\ln(e^x-1)} = e^x - 1$$

$$e^{\ln(e^x-1)} \cdot (\ln(e^x-1))' = e^x$$

$$(\ln(e^x-1))' = \frac{e^x}{e^x-1}$$

3-9 Inverse Trigonometric Functions

師大工教一

Definition:

1. $y = \sin^{-1} x$ is the number in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ for which $\sin y = x$.
2. $y = \cos^{-1} x$ is the number in $[0, \pi]$ for which $\cos y = x$.
3. $y = \tan^{-1} x$ is the number in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which $\tan y = x$.
4. $y = \cot^{-1} x$ is the number in $(0, \pi)$ for which $\cot y = x$.
5. $y = \sec^{-1} x$ is the number in $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$ for which $\sec y = x$.
6. $y = \csc^{-1} x$ is the number in $\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$ for which $\csc y = x$.

The Derivative of $y = \sin^{-1} x$

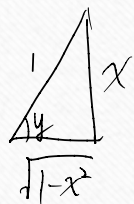
Derive $(\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}.$

$$y = \sin^{-1} x$$

$$\sin y = \sin(\sin^{-1} x) = x$$

$$\cos y \cdot y' = 1$$

$$y' = \frac{1}{\cos y}$$



$$y' = \frac{1}{\sqrt{1-x^2}}$$

Chain Rule: $\left(\sin^{-1}(f(x))\right)' = \frac{1}{\sqrt{1-(f(x))^2}} \cdot f'(x).$

Ex2: Find $\frac{d}{dx}(\sin^{-1} x^2).$

$$= \frac{1}{\sqrt{1-x^4}} \cdot 2x = \frac{2x}{\sqrt{1-x^4}}$$

The Derivative of $y = \tan^{-1} x$

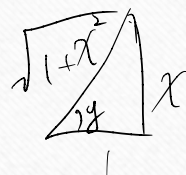
Derive $(\tan^{-1} x)' = \frac{1}{1+x^2}$.

$$y = \tan^{-1} x$$
$$\tan y = x$$

$$y' = \frac{1}{1+x^2}$$

$$\sec^2 y \cdot y' = 1$$

$$y' = \frac{1}{\sec^2 y}$$



Chain Rule: $(\tan^{-1}(f(x)))' = \frac{1}{1+(f(x))^2} \cdot f'(x)$.

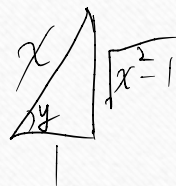
The Derivative of $y = \sec^{-1} x$

Derive $(\sec^{-1} x)' = \frac{1}{|x|\sqrt{x^2-1}}$.

$$\sec y = x = \frac{1}{\cos y}$$

$$\sec y \cdot \tan y \cdot y' = 1$$

$$y' = \frac{1}{\sec y \cdot \tan y} = \begin{cases} \frac{1}{x\sqrt{x^2-1}}, & x > 0 \\ \frac{1}{-x\sqrt{x^2-1}}, & x < 0 \end{cases} = \frac{1}{|x|\sqrt{x^2-1}}$$



$$\sec^{-1} x, \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$$

inverse function 要有 one to one
∴ 只討論 $x > 0$ 的情況

Chain Rule: $\left(\sec^{-1}(f(x))\right)' = \frac{1}{|f(x)|\sqrt{f(x)^2 - 1}} \cdot f'(x).$

Ex3: Find $\frac{d}{dx}(\sec^{-1}(5x^4)).$

$$= \frac{1}{|5x^4|\sqrt{25x^8 - 1}} \cdot 20x^3 = \frac{20x^3}{|5x^4|\sqrt{25x^8 - 1}}$$

Inverse Function-Inverse Cofunction Identities

$$1. \cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

$$2. \cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$$

$$3. \csc^{-1} x = \frac{\pi}{2} - \sec^{-1} x$$

Chain Rule:

$$1. \left(\cos^{-1}(f(x)) \right)' = \frac{-1}{\sqrt{1-(f(x))^2}} \cdot f'(x)$$

$$2. \left(\cot^{-1}(f(x)) \right)' = \frac{-1}{1+(f(x))^2} \cdot f'(x)$$

$$3. \left(\csc^{-1}(f(x)) \right)' = \frac{-1}{|f(x)|\sqrt{f(x)^2-1}} \cdot f'(x)$$

HW3-9

- **HW: 10,13,15,21,31,33**

105分 3. (14 points) Find the derivatives $\frac{dy}{dx}$

(a) $y = e^{(\ln|5x| + (\cos x)^2)^2}, \quad x \neq 0,$

V (b) $y = \arctan(\sqrt{1-x^2}), \quad |x| < 1.$

$$(b) \quad y' = \frac{1}{1+(1-x^2)} \cdot \left[(1-x^2)^{\frac{1}{2}} \right]' = \frac{1}{1-x^2} \left(\frac{1}{2} (1-x^2)^{-\frac{1}{2}} \cdot (-2x) \right) = \frac{-1}{1-x^2} \cdot \frac{1}{\sqrt{1-x^2}} \cdot x = \frac{-x}{(1-x^2)\sqrt{1-x^2}}$$

108 2. (24 points) Find the derivatives $\frac{dy}{dx}$

(a) $y = \ln(2x + \sqrt{x} + 1), \quad x > 0,$

V (b) $y = \cos(\arctan e^{3x}), \quad (\text{note : } \arctan x = \tan^{-1} x)$

V (c) $y = x^{2/x}, \quad x > 0.$

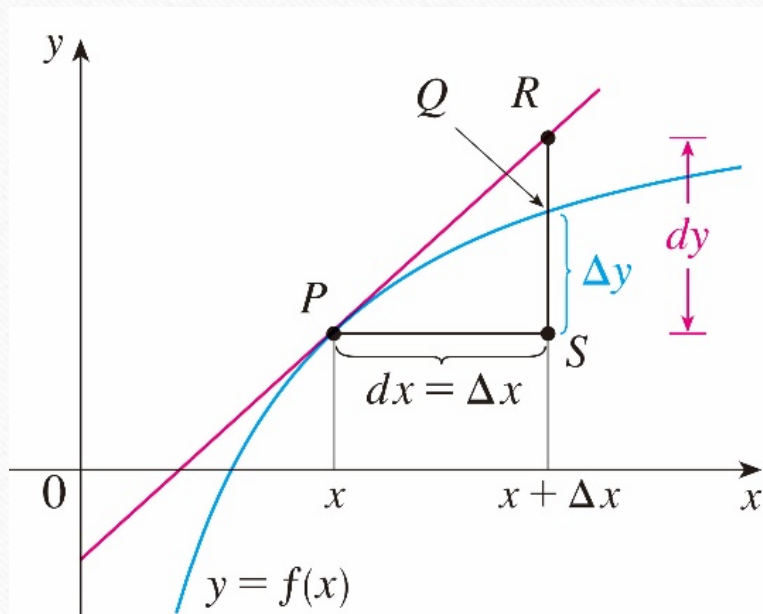
$$(b) \quad y' = -\sin(\arctan e^{3x}) \cdot \frac{1}{1+e^{6x}} \cdot (e^{3x})' \\ = -\sin(\arctan e^{3x}) \cdot \frac{3e^{3x}}{1+e^{6x}} = \frac{-3e^{3x} \sin(\arctan e^{3x})}{1+e^{6x}}$$

$$(c) \quad y' = \left(x^{\frac{2}{x}} \right)' \\ \ln y = \frac{2}{x} \ln x \\ \frac{y'}{y} = \frac{2 \cdot \frac{1}{x} \cdot x - 2 \ln x \cdot 1}{x^2} \\ y' = x^{\frac{2}{x}} \left[\frac{2-2\ln x}{x^2} \right]$$

Differentials

不求

Definition: Let $y = f(x)$ be a differentiable function. The differential dx is an independent variable. The differential dy is $dy = f'(x)dx$.



Ex4(p214)

(a) Find dy if $y = x^5 + 37x$.

(b) Find the value of dy when $x = 1$ and $dx = 0.2$.

$$\begin{aligned} (a) \quad dy &= f'(x) dx \\ &= (5x^4 - 37) dx \end{aligned}$$

$$(b) \quad dy \Big|_{\substack{x=1, \\ dx=0.2}} = (5 - 37) \times 0.2 = -32 \times 0.2 = -6.4$$